Given that, the utility function $U = f(q_1, q_2)$

and the budget constraint $y^0 = P_1 q_1 + P_2 q_2$

The utility maximization problem can be written as:

$$V = f(q_1, q_2) + \lambda(y^0 - P_1 q_1 - P_2 q_2) \quad \cdots (1)$$

The first-order condition implies that the partial derivatives of (1) with respect to q_1, q_2, and λ equal to zero:

$$\left.\begin{aligned}
\frac{\partial V}{\partial q_1} &= f_1 - \lambda P_1 = 0 \quad \text{or, } f_1 = \lambda P_1 \quad \text{or, } \lambda = \frac{f_1}{P_1} \\
\frac{\partial V}{\partial q_2} &= f_2 - \lambda P_2 = 0 \quad \text{or, } f_2 = \lambda P_2 \quad \text{or, } \lambda = \frac{f_2}{P_2} \\
\frac{\partial V}{\partial \lambda} &= y^0 - P_1 q_1 - P_2 q_2 = 0
\end{aligned}\right\} (2)$$

From the first-order condition we can write $\frac{f_1}{P_1} = \frac{f_2}{P_2} = \lambda$

$$\text{or, } \frac{f_1}{f_2} = \frac{P_1}{P_2} = \lambda \quad \cdots (3)$$

In order to find the magnitude of the effect of price and income changes on the consumer's purchases, allow all variables to vary simultaneously. This is accomplished by total differentiation of equation (2):

$$f_{11} dq_1 + f_{12} dq_2 - P_1 d\lambda - \lambda dP_1 = 0$$

$$f_{21} dq_1 + f_{22} dq_2 - P_2 d\lambda - \lambda dP_2 = 0$$

$$dy - P_1 dq_1 - q_1 dP_1 - P_2 dq_2 - q_2 dP_2 = 0$$

or, we can write

$$\left.\begin{aligned}
f_{11} dq_1 + f_{12} dq_2 - P_1 d\lambda &= \lambda dP_1 \\
f_{12} dq_2 + f_{22} dq_2 - P_2 d\lambda &= \lambda dP_2 \\
-P_1 dq_1 - P_2 dq_2 \qquad &= -dy + q_1 dP_1 + q_2 dP_2
\end{aligned}\right\} (4)$$

In order to solve this system of three equations for the three unknowns dq_1, dq_2, and $d\lambda$, the three terms on the right must be regarded as constants.

Now in matrix form we can write:

$$\begin{bmatrix} f_{11} & f_{12} & -P_1 \\ f_{21} & f_{22} & -P_2 \\ -P_1 & -P_2 & 0 \end{bmatrix} \begin{bmatrix} dq_1 \\ dq_2 \\ d\lambda \end{bmatrix} = \begin{bmatrix} \lambda dP_1 \\ \lambda dP_2 \\ -dy + q_1 dP_1 + q_2 dP_2 \end{bmatrix} \quad \ldots (5)$$

Denoting the determinant by D and the cofactor of the element in the first row and the first column by D_{11}, the cofactor of the element in the first row and second column by D_{12}, etc., the solution of (5) by Cramer's rule is:

$$dq_1 = \frac{\begin{vmatrix} \lambda dP_1 & f_{12} & -P_1 \\ \lambda dP_2 & f_{22} & -P_2 \\ -dy + q_1 dP_1 + q_2 dP_2 & -P_2 & 0 \end{vmatrix}}{D}$$

$$= \frac{\lambda D_{11} dP_1 - \lambda D_{21} dP_2 + D_{31}(-dy + q_1 dP_1 + q_2 dP_2)}{D} \quad (6)$$

$$dq_2 = \frac{\begin{vmatrix} f_{11} & \lambda dP_1 & -P_1 \\ f_{21} & \lambda dP_2 & -P_2 \\ -P_1 & -dy + q_1 dP_1 + q_2 dP_2 & 0 \end{vmatrix}}{D}$$

$$= \frac{-\lambda D_{12} dP_1 + \lambda D_{22} dP_2 - D_{32}(-dy + q_1 dP_1 + q_2 dP_2)}{D} \quad -(7)$$

and $$d\lambda = \frac{\lambda D_{13} dP_1 - \lambda D_{23} dP_2 + D_{33}(-dy + q_1 dP_1 + q_2 dP_2)}{D} \quad -(8)$$

Now, we want to find out the change of q_1 with respect to change in P_1. Therefore, we assume all other things (P_2 and y) remain constant, so $dP_2 = dy = 0$. Substituting $dP_2 = dy = 0$ into equ. (6) we get

$$\partial q_1 = \frac{\lambda D_{11}\, \partial P_1 + 0 + D_{31}(q_1 \partial P_1)}{D}$$

$$\text{or, } \partial q_1 = \frac{\lambda D_{11}\, \partial P_1 + q_1 D_{31}\, \partial P_1}{D}$$

$$\text{or, } \partial q_1 = \frac{\partial P_1 (\lambda D_{11} + q_1 D_{31})}{D}$$

$$\text{or, } \frac{\partial q_1}{\partial P_1} = \frac{\lambda D_{11}}{D} + q_1 \frac{D_{31}}{D} \quad \cdots (9)$$

The partial derivative on the left-hand side of equ. (9) is the rate of change of the consumer's purchase of Q_1 with respect to ~~Q_1~~ changes in P_1, all other things being equal.

Now, ~~of~~ ~~w~~ we can find out the impact of relative income change on q_1. When all other things being equal (P_1 and P_2 constant so $dP_1 = dP_2 = 0$), the rate of change with respect to income is: $dq_1 = \frac{0 - 0 - dy\, D_{31}}{D}$, so,

$$\frac{\partial q_1}{\partial y} = -\frac{D_{31}}{D} \quad \cdots (10)$$

Now, consider a price change that is compensated by an income change that leaves the consumer on his indifference curve. An increase in the price of a commodity is accompanied by a corresponding increase in his income such that $dU = 0$ and $f_1 dq_1 + f_2 dq_2 = 0$ $[\because U = f(q_1, q_2)]$

From equation (3); since $\frac{f_1}{f_2} = \frac{P_1}{P_2}$; it is also true that $P_1 dq_1 + P_2 dq_2 = 0$. Hence, from the last equation of (4), $-dy + q_1 dP_1 + q_2 dP_2 = 0$; and, $dy = 0$. Then we can write (6) as

$$\partial q_1 = \frac{\lambda D_{11} \partial P_1 - 0 + 0}{D}$$

$$\text{or, } \left(\frac{\partial q_1}{\partial P_1}\right)_{U=\text{constant}} = \frac{\lambda D_{11}}{D} \quad \cdots (11)$$

So, we can write equ. (9) as

$$\frac{\partial q_1}{\partial P_1} = \frac{D_{11}\lambda}{D} + q_1 \frac{D_{31}}{D}$$

$$\text{or, } \frac{\partial q_1}{\partial P_1} = \left(\frac{\partial q_1}{\partial P_1}\right)_{U=\text{constant}} - q_1 \left(\frac{\partial q_1}{\partial y}\right)_{\text{Prices}=\text{Constant}} \quad \cdots (12)$$

Equation (12) is known as the Slutsky equation. The $\frac{\partial q_1}{\partial P_1}$ is the slope of the ordinary demand curve for q_1 and the first term on the right hand side is the slope of the compensated demand curve for q_1.